

Q. MVE Assignment #2

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Q#1.

$$C(x, y) = e^x + \sin y - 4x - y$$

$$C(x, y) = e^x + \sin y - 4x - y$$

$$\frac{d}{dx} = e^x - 4$$

$$f_x = 0 \\ e^x - 4 = 0 \\ e^x = 4$$

$$\frac{d}{dy} = \cos y - 1$$

$$f_y = 0 \\ \cos y - 1 = 0 \\ \cos y = 1$$

$$e^x = 4$$

Taking $\ln$ B.S

$$e^{\ln x} = \ln 4$$

$$x = \ln 4$$

$$\cos y = 1$$

$$y = \cos^{-1}(1)$$

$$y = 0 \text{ or } 2\pi$$

=> Critical point:- $(\ln 4, 2\pi)$

$$f_{xx} = e^x \\ f_{yy} = -\sin y$$

$$f_{xy} = 0$$

$$D = (f_{xx})(f_{yy}) - (f_{xy})^2 \\ (e^x)(-\sin y) = 0$$

$$= -e^x \sin y$$

$$x = \ln 4 \\ y = 2\pi$$

$$= -e^{\ln 4} \sin(2\pi) = -4 \cdot 0 = 0$$

Incorrect

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Q#2: $m(a,b) = 100a + 150b - 2a^2 - 3b^2 - ab$

$m(a,b) = 100a + 150b - 2a^2 - 3b^2 - ab$

$$\frac{dm}{da} = 100 - 4a - b$$

$$\frac{dm}{db} = 150 - 6b - a$$

$$f_x = 0$$

$$100 - 4a - b = 0$$

$$f_y = 0$$

$$150 - 6b - a = 0$$

$$-a = 150 + 6b$$

$$100 - 4(150 + 6b) - b = 0$$

$$+a = 150 + 6b$$

$$100 - 600 + 24b - b = 0$$

$$-a = 150 - 6b$$

$$-500 + 23b = 0$$

$$23b = 500$$

$$b = \frac{500}{23}$$

$$a = 150 - 6\left(\frac{500}{23}\right)$$

$$a = \frac{3450 - 3000}{23}$$

$$a = \frac{450}{23}$$

$\Rightarrow$ Critical points: $\frac{450}{23}, \frac{500}{23}$

$$f_{xx} = -4$$

$$f_{yy} = -6$$

$$f_{xy} = -1$$

$$D = (f_{xx})(f_{yy}) - (f_{xy})^2$$

$$= (-4)(-6) - (-1)^2$$

$$= 24 - 1$$

$$= 23$$

$D > 0, f_{xx} < 0$
(local maximum)

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Q43: $f(m,n) = 3m^2 + 2mn + 4n^2$

soln. $g(m,n) = m^2 + n^2 + mn = 400$

$$\frac{df}{dm} = 6m + 2n$$

$$\frac{dg}{dm} = 2m + n$$

$$\frac{df}{dn} = 2m + 8n$$

$$\frac{dg}{dn} = 2n + m$$

$$f_x = g_x$$

$$f_y = g_y$$

$$6m + 2n = (2m + n)\lambda$$

$$(2m + 8n) = (2n + m)\lambda$$

$$\cancel{2m + 8n = 2m + m\lambda}$$

$$\frac{6m + 2n}{2m + n} = \lambda$$

$$\frac{2m + 8n}{2n + m} = \lambda$$

$$\frac{6m + 2n}{2m + n} = \frac{2m + 8n}{2n + m}$$

$$(2n + m)(6m + 2n) = (2m + n)(2m + 8n)$$

$$(2n)(6m) + (m)(6m) + (2n)(2n) + (2n)(m) = (2m)(8n) + (2m)(2m) + n(2m) + n(8n)$$

$$12mn + 6m^2 + 4n^2 + 2nm = 16mn + 4m^2 + 2mn + 8n^2$$

$$14mn + 6m^2 + 4n^2 = 18mn + 4m^2 + 8n^2$$

$$= 2m^2 - 4mn - 4n^2 = 0$$

$$2(m^2 - 2mn - 2n^2) = 0$$

$$m^2 - 2mn - 2n^2$$

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2) Use quadratic formula:-

$$m = 2n \pm \sqrt{(2n)^2 - (4)(1)(-2n)}$$

2(1)

$$m = \frac{2n \pm \sqrt{4n^2 + 8n}}{2} \Rightarrow \frac{2n \pm \sqrt{4n^2}}{2} = \frac{2n \pm 2n\sqrt{3}}{2}$$

$$m = \frac{2n(1 \pm \sqrt{3})}{2} = n(1 \pm \sqrt{3})$$
$$[m = n(1 + \sqrt{3})]$$

Now put in constraint:-

$$m = n(1 + \sqrt{3}) \text{ in } m^2 + n^2 + nm - 400 = 0$$

$$\Rightarrow (n(1 + \sqrt{3}))^2 + n^2 + n(n(1 + \sqrt{3})) - 400 = 0$$

$$\Rightarrow n^2(1 + \sqrt{3})^2 + n^2 + n^2(1 + \sqrt{3}) - 400 = 0$$

$$\Rightarrow n^2(4 + 2\sqrt{3}) + n^2 + n^2(1 + \sqrt{3}) = 400$$

$$\Rightarrow n^2(6 + 3\sqrt{3}) = 400$$

$$n^2 = \frac{400}{6 + 3\sqrt{3}} \Rightarrow \sqrt{n^2} = \sqrt{\frac{400}{6 + 3\sqrt{3}}}, n = \frac{20}{\sqrt{6 + 3\sqrt{3}}} \approx 5.977$$

$$m = 16.34$$

$$\text{Or } n = -5.977, m = -16.32$$

$$(16.34, 5.977) \text{ or } (-16.32, -5.977)$$

2) Values of (m, n) are $(16.34, 5.977)$, $(-16.34, -5.977)$, $(22.307, -16.32)$ and $(-22.307, 16.32)$ pm

(m, n)	$C(m, n)$	minimize/maximize
$(16.34, 5.977)$	1139.5120	maximum
$(-16.34, -5.977)$	1139.20	maximum
$(22.307, -16.32)$	35.84420	minimize
$(-22.307, 16.32)$	35.84420	maximum

or

$(16.32, 5.977)$ the minimize value

Q#4:- $E(p, q, r) = p^2 + 4q^2 + 2r^2$ $p^2 + q^2 + r^2 = 250,000$

SOL:-
 $E_p = \frac{\partial E}{\partial p} = 2p$, $E_q = \frac{\partial E}{\partial q} = 8q$, $E_r = \frac{\partial E}{\partial r} = 4r$
 $G_p = \frac{\partial G}{\partial p} = 2p$, $G_q = \frac{\partial G}{\partial q} = 2q$, $G_r = \frac{\partial G}{\partial r} = 2r$

$f_1 = \lambda G_1$ $f_2 = \lambda G_2$ $f_3 = \lambda G_3$

$\frac{\partial E}{\partial p} = \lambda \frac{\partial G}{\partial p}$, $\frac{\partial E}{\partial q} = \lambda \frac{\partial G}{\partial q}$, $\frac{\partial E}{\partial r} = \lambda \frac{\partial G}{\partial r}$
 $\lambda = 1$, $\lambda = 4$, $\lambda = 2$

Inconsistent equation:-

Now consider when one or more variables are zero.

Case 1: $q=0, r=0$ the constraint becomes:
 $p^2 = \sqrt{250000}$ $p=500$
 $E(p, q, r) = 250000$

Case 2: $p=0, r=0$
 $q^2 = \sqrt{250,000}$ $q=500$
 $E(p, q, r) = 1000000$

Case 3: $p=0, q=0$
 $r^2 = \sqrt{250,000}$ $r=500$ $E(p, q, r) = 500000$

∴ The minimum energy consumption occurs in case 1 when $p=500, q=0, r=0$ Ans.

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Q45: $J(\theta_1, \theta_2) = \ln(\theta_1) + \theta_1^2 + \theta_1 \theta_2$
initial values (1,1) and $\alpha = 0.1$
Sol:

$$J_{\theta_1} = \cos(\theta_1) + \theta_2$$

$$J_{\theta_2} = 2\theta_2 + \theta_1$$

1st iteration:-

$$J_{\theta_1}(1,1) = \cos(1) + 1 \Rightarrow 0.999 + 1 \Rightarrow 1.999$$

$$J_{\theta_2}(1,1) = 2(1) + (1) \Rightarrow 3$$

update the value of θ_1 & θ_2 .

$$x = x_0 - \alpha \cdot \frac{df}{dx}$$

$$y = y_0 - \alpha \cdot \frac{df}{dy}$$

$$\theta_1 = 1 - 0.1 \times 1.999$$

$$\theta_2 = 1 - 0.1 \times 3$$

$$\theta_1 = 0.8001$$

$$\theta_2 = 0.7$$

$J(0.8001, 0.7) = 1.06$, the value of function at 1st iteration.

2nd iteration:-

$$J_{\theta_1} = \cos(0.8001) + 0.7 = 1.699$$

$$J_{\theta_2} = 2(0.7) + 0.8001 = 2.2001$$

updated values:-

$$x_2 = x_1 - \alpha \cdot \frac{df}{dx}$$

$$y_2 = y_1 - \alpha \cdot \frac{df}{dy}$$

$$\theta_1 = 0.8001 - 0.1 \times 1.699$$

$$\theta_2 = 0.7 - 0.1 \times 2.2001$$

$$\theta_1 = 0.6302$$

$$\theta_2 = 0.4799$$

$J(0.6302, 0.4799) = 0.5437$ at iteration 2.

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Q#6 - $R(f, g, h) = f^2 + 2g^2 + 3h^2 + f - 7g - 9h + 30$

$\alpha = 0.05$ initial value (1,1,1)

$R_f = 2f + 1$, $R_g = 4g - 7$, $R_h = 6h - 9$

1st iteration

$R_f(1,1,1) = 2 + 1 = 3$

$R_g(1,1,1) = 4 - 7 = -3$

$R_h(1,1,1) = 6 - 9 = -3$

update values of f, g & h :

$f_1 = f_0 - \frac{\alpha \cdot R_f}{R_f}$, $g_1 = g_0 - \frac{\alpha \cdot R_g}{R_g}$, $h_1 = h_0 - \frac{\alpha \cdot R_h}{R_h}$

$f_1 = 1 - 0.05 \times 3 = 0.85$

$g_1 = 1 - 0.05 \times (-3) = 1.15$

$h_1 = 1 - 0.05 \times (-3) = 1.15$

$R_f(0.85, 1.15, 1.15) = -1.7$

$R_g(0.85, 1.15, 1.15) = -1.5$

$R_h(0.85, 1.15, 1.15) = -2.1$

$f_2 = f_1 - \frac{\alpha \cdot R_f}{R_f}$

$g_2 = g_1 - \frac{\alpha \cdot R_g}{R_g}$

$h_2 = h_1 - \frac{\alpha \cdot R_h}{R_h}$

$f_2 = 0.85 - 0.05 \times (-1.7) = 1.185$

$g_2 = 1.15 - 0.05 \times (-1.5) = 1.175$

$h_2 = 1.15 - 0.05 \times (-2.1) = 1.255$

$R(1.185, 1.175, 1.255)$ at

second iteration.

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Q#7.

$$J(\theta_1, \theta_2, \theta_3) = e^{\theta_1} + \theta_2 + 4\theta_3^2 + \theta_1\theta_2$$

$$\frac{\partial J}{\partial \theta_1} = e^{\theta_1} + \theta_2, \quad \frac{\partial J}{\partial \theta_2} = 2\theta_2 + \theta_1$$

$$\frac{\partial J}{\partial \theta_3} = 8\theta_3$$

2) find C.P.

$$e^{\theta_1} + \theta_2 = 0 \quad (1)$$

$$2\theta_2 + \theta_1 = 0 \quad (2)$$

$$8\theta_3 = 0 \quad (3)$$

$$\theta_3 = 0 \rightarrow \text{from eq. 3}$$

$$\theta_1 = -2\theta_2 \rightarrow \text{from eq. 2}$$

$$\text{Substitute } \theta_1 = -2\theta_2$$

$$-2\theta_2$$

$$e + \theta_2 = 0$$

$$f(\theta_2) = e^{-2\theta_2} + \theta_2 \quad \text{to find } f(\theta_2) = 0$$

$$[\theta_2 = -0.567] \text{ then}$$

$$\theta_1 = -2\theta_2 \quad \text{put } \theta_2 \text{ value}$$

$$[\theta_1 = 1.134]$$

So, CP are

$$\theta_1 = 1.134$$

$$\theta_2 = -0.167$$

$$\theta_3 = 0$$

}
Optimal values

New Hessian matrix:

$$H = \begin{bmatrix} \frac{\partial^2 J}{\partial \theta_1^2} & \frac{\partial^2 J}{\partial \theta_1 \partial \theta_2} & \frac{\partial^2 J}{\partial \theta_1 \partial \theta_3} \\ \frac{\partial^2 J}{\partial \theta_2 \partial \theta_1} & \frac{\partial^2 J}{\partial \theta_2^2} & \frac{\partial^2 J}{\partial \theta_2 \partial \theta_3} \\ \frac{\partial^2 J}{\partial \theta_3 \partial \theta_1} & \frac{\partial^2 J}{\partial \theta_3 \partial \theta_2} & \frac{\partial^2 J}{\partial \theta_3^2} \end{bmatrix}$$

Now

$$\frac{\partial^2 J}{\partial \theta_1^2} = 2$$

$$\frac{\partial^2 J}{\partial \theta_1 \partial \theta_2} = 1$$

$$\frac{\partial^2 J}{\partial \theta_1 \partial \theta_3} = 0$$

$$\frac{\partial^2 J}{\partial \theta_2 \partial \theta_1} = 1$$

$$\frac{\partial^2 J}{\partial \theta_2^2} = 2$$

$$\frac{\partial^2 J}{\partial \theta_2 \partial \theta_3} = 0$$

$$\frac{\partial^2 J}{\partial \theta_3 \partial \theta_1} = 0$$

$$\frac{\partial^2 J}{\partial \theta_3 \partial \theta_2} = 0$$

$$\frac{\partial^2 J}{\partial \theta_3^2} = 8$$

$$H = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 8 \end{bmatrix}$$

$H =$

$$H = \begin{bmatrix} 3.108 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 8 \end{bmatrix}$$

$$H = 3.108 > 0$$

$$H = 5.216 > 0$$

$$H = 41.728 > 0$$

}
all positive
definite

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Q#8:-

$$I = \iint_R xy \cdot x^2 dA$$

$$x \in (0, 1) \\ y \in (1, 3)$$

Soln

$$I = \int_0^1 \int_1^3 xy \cdot x^2 dy dx$$

$$\Rightarrow \int_1^3 xy \cdot x^2 dy \Rightarrow x \int_1^3 (y \cdot x) dy \Rightarrow \left[\frac{y^2}{2} \cdot xy \right]_1^3$$

$$\Rightarrow \frac{9}{2} - 3x - \left(\frac{1}{2} - x \right) \Rightarrow \frac{9}{2} - 3x - \frac{1}{2} + x$$

$$\Rightarrow \frac{9 - 6x - 1 + 2x}{2} = \frac{8 - 4x}{2} \Rightarrow \cancel{x} \frac{4 - 2x}{x}$$

$$\Rightarrow \int_0^1 x(4 - 2x) dx$$

$$\Rightarrow \int_0^1 4x - 2x^2 dx \Rightarrow \left[\frac{4x^2}{2} - \frac{2x^3}{3} \right]_0^1$$

$$\Rightarrow 2(1)^2 - \frac{2(1)^3}{3} - \left(2(0)^2 - \frac{2(0)^3}{3} \right)$$

$$\Rightarrow 2 - \frac{2}{3} = \frac{6-2}{3} = \frac{4}{3}$$

$$\boxed{I = \frac{4}{3}} \text{ Ans}$$

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Q49

$$z = 10 - x^2 - y^2$$

vertices of triangle (0,0), (2,0)
and (2,3)

find the volume of the mountain above
the xy-plane.

first find a & b wrt x & y points.

This x-axis (y,0).

vertical line x=2.

The line connecting (0,0) & (2,3)

$$m = \frac{3-0}{2-0} = \frac{3}{2} \quad y = \frac{3x}{2}$$

∴ The limits of integration are:

$$x: 0 \leq x \leq 2$$

$$y: 0 \leq y \leq \frac{3x}{2}$$

$$V = \iint (10 - x^2 - y^2) dy dx$$

$$V = \int_0^2 \int_0^{\frac{3x}{2}} (10 - x^2 - y^2) dy dx$$

$$\int_0^{\frac{3x}{2}} (10 - x^2 - y^2) dy = \left[10y - x^2y - \frac{y^3}{3} \right]_0^{\frac{3x}{2}}$$

$$10\left(\frac{3x}{2}\right) - x^2\left(\frac{3x}{2}\right) - \frac{\left(\frac{3x}{2}\right)^3}{3} - \left(10(0) - x^2(0) - \frac{0}{3}\right)$$

$$= 15x - \frac{3x^3}{2} - \frac{9x^3}{8}$$

$$= 15x - \frac{11x^3}{8}$$

$$V = \int_0^2 \left(\frac{15x}{8} - \frac{21x^3}{8} \right) dx$$

$$V = \left[\frac{15x^2}{2 \cdot 8} - \frac{21x^4}{4 \cdot 8} \right]_0^2$$

$$V = \frac{15(2)^2}{2 \cdot 8} - \frac{21(2)^4}{4 \cdot 8} = \left(\frac{15(4)}{16} - \frac{21(16)}{32} \right)$$

$$V = \frac{30 - 21}{2} = \frac{9}{2} = 4.5 \text{ cubic units}$$

Ans

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Q.1101

(a) $f(x, y, z) = x^2 + y^2 - z$ at $(1, 2, 5)$.

Using: $f_x(x-x_0) + f_y(y-y_0) + f_z(z-z_0)$ for tangent plane:

$$\frac{df}{dx} = 2x \quad \frac{df}{dy} = 2y \quad \frac{df}{dz} = -1$$

$$\begin{aligned} f_x &= 2(1) = 2 & (2, 4, -1) \\ f_y &= 2(2) = 4 \\ f_z &= -1 \end{aligned}$$

$$= 2(x-1) + 4(y-2) + (-1)(z-5)$$

$$2x - 2 + 4y - 8 - z + 5$$

$$2x + 4y - z - 5 = 0$$

$[2x + 4y - z - 5] \rightarrow$ tangent line equation.

$\Rightarrow$ Normal line equation

$$x = 1 + 2t$$

$$z = 5 - t$$

$$y = 2 + 4t$$

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(b) $f_1(x, y, z) = x^2 + y^2 + z^2 - 9$, $f_2(x, y, z) = x + 2y - 3z + 4$

at intersection point $(2, -1, 1)$.

Soln:

$$f_{1x} = \frac{\partial}{\partial x} \Rightarrow 2(2) = 4 \quad (4, -2, 2)$$

$$f_{1y} = \frac{\partial}{\partial y} \Rightarrow 2(-1) = -2$$

$$f_{1z} = \frac{\partial}{\partial z} \Rightarrow 2(1) = 2$$

$$f_{2x} = 1$$

$$f_{2y} = 2 \quad (1, 2, -3)$$

$$f_{2z} = -3$$

$$\Rightarrow \cos \theta = \frac{(4, -2, 2) \cdot (1, 2, -3)}{\| (4, -2, 2) \| \| (1, 2, -3) \|} = \frac{4 - 4 - 6}{2\sqrt{6} \times \sqrt{14}} = -\frac{3}{\sqrt{21}}$$

$$\frac{-6}{2\sqrt{84}} \Rightarrow \frac{-3}{\sqrt{21}} \quad \Rightarrow \frac{-3}{\sqrt{21}} \times \frac{\sqrt{21}}{\sqrt{21}} = \frac{-3\sqrt{21}}{\sqrt{21}}$$

$$\cos \theta = \frac{-\sqrt{21}}{14} \quad \Rightarrow \quad \theta = \cos^{-1}\left(\frac{-\sqrt{21}}{14}\right) = 109.10^\circ$$

ans